

Multiple Choice

1. **Answer:** A

Options: A) Mimetic coloration; B) Aposomatic coloration; C) Deceptive markings; D) Batesian mimicry; E) Flash coloration

Note: Mimetic coloration is correct because it refers to the adaptation of organisms to resemble their surroundings, enabling them to avoid detection by predators.

2. **Answer:** A

Options: A) Tonus refers to the electrical activity in the brain during relaxation.; B) Tonus refers to the maximum force a muscle can exert in a single effort.; C) Tonus refers to the flexibility and range of motion in joints and muscles.; D) Tonus refers to the state of complete relaxation in skeletal muscles.; E) Tonus refers to the involuntary contraction of smooth muscles in the digestive tract.

Note: The term tonus, or tone, refers to the baseline level of electrical activity in the brain that occurs during states of relaxation, indicating a state of readiness and responsiveness of the nervous system.

3. **Answer:** A

Options: A) Courtship is primarily for entertainment purposes among terns; B) Courtship is not necessary for reproduction; C) Courtship is a way to show dominance; D) Courtship is a random behavior with no impact on reproduction; E) Courtship is necessary for physical transformation required for reproduction

Note: The correct answer highlights that courtship behaviors, such as the male presenting a fish to a female tern, serve not only to strengthen pair bonds and ensure reproductive success but also provide a means of entertainment, which can enhance social interactions and cohesion within the species.

4. **Answer:** C

Options: A) conjugation results in no genetic diversity; B) conjugation requires more than two organisms; C) all individuals are identical genetically, and conjugation could not occur; D) repeated fission in paramecia results in sterile offspring that cannot conjugate; E) the group of paramecia lacks the cellular structures necessary for conjugation

Note: Conjugation in paramecia requires genetic diversity for the exchange of genetic material, and a group descended from a single individual through repeated fission would be genetically identical, making conjugation impossible.

5. **Answer:** A

Options: A) peroxisome; B) cytoplasm; C) nucleus; D) intermembrane space; E) outer mitochondrial membrane

Note: The Krebs cycle, also known as the citric acid cycle, actually occurs in the mitochondria of human cells, not the peroxisome, as it is the primary pathway for

aerobic respiration and energy production, highlighting a misunderstanding in the provided answer.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Lysosomes and endosomes are similar in size and density, making them difficult to separate by low-speed centrifugation.

7. **Answer:** False

Options: A) True; B) False

Note: The Hardy-Weinberg law describes a theoretical genetic equilibrium in a population where allele frequencies remain constant over generations in the absence of evolutionary influences, indicating that genetic diversity can be maintained without gene exchange with other populations.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because if meiosis produces cells with 32 chromosomes, that would indicate a haploid number of 32, implying a diploid number of 64, rather than half the diploid number as suggested.

9. **Answer:** False

Options: A) True; B) False

Note: Crossover primarily occurs between homologous genes located on the same chromosome, as this is necessary for the exchange of genetic material during meiosis, rather than between genes on different chromosomes.

10. **Answer:** True

Options: A) True; B) False

Note: In eukaryotic photosynthetic cells, the electrons energized by light in photosystem I ultimately reduce NADP⁺ to NADPH, which is essential for the Calvin cycle in photosynthesis.

Long Form Response

11. **Answer:** Certain animals, such as the robin, are endothermic or warm-blooded, allowing them to maintain a constant body temperature despite environmental fluctuations. This ability is primarily achieved through metabolic processes that generate heat, along with physiological adaptations like insulation through feathers, which minimize heat loss. In contrast, ectothermic animals like lizards rely on behavioral adaptations, such as basking in the sun to absorb heat or seeking shade to prevent overheating. These mechanisms highlight the diverse strategies animals employ to regulate their body temperature, illustrating the evolutionary significance of thermoregulation for survival and

activity in varying habitats.

Note: correct because it accurately distinguishes between endothermic and ectothermic animals, explaining how metabolic processes and physiological adaptations, such as insulation and behavioral strategies, enable these animals to regulate their body temperature in response to environmental changes.

12. **Answer:** Bacteria play a vital role in the dairy industry, primarily through their involvement in the production of fermented milk products such as yogurt, kefir, and various cheeses. These microorganisms are essential for fermentation processes, as they convert lactose into lactic acid, which not only enhances the flavor and texture of the products but also extends their shelf life. Additionally, certain milk-giving animals rely on bacteria to aid in nutrient acquisition; these bacteria help break down cellulose from the plant material consumed by the animals, facilitating the digestion of fibrous foods and improving nutrient absorption. Overall, the symbiotic relationship between bacteria and dairy production underscores the importance of these microorganisms in both food processing and animal nutrition.

Note: correct because it accurately describes how bacteria facilitate fermentation in dairy products, enhancing flavor, texture, and shelf life, while also emphasizing their role in nutrient acquisition for milk-giving animals through the breakdown of lactose.